SUBJECT: LOGICAL REASONING-OE

Assignment No. 1: FUNDAMENTAL OF LOGICAL REASONING. DOS: 20/07/2026

1. In a certain code language, if the word 'MACHINE' is coded as 'LBBGJOF', what is the numerical equivalent of the coded word when letters A-Z are mapped to 1-26 and summed together? Find the final sum
2. In a number series: 4, 9, 19, 39, 79, X, what is the exact numerical value of X obtained by following the established arithmetic progression and multiplier pattern?
3. If a clock shows that the sum of the digits representing the current hours and minutes on a 24-hour digital clock format reads 22, and 4 hours and 15 minutes are added to it, what will be the new digital time represented in total elapsed minutes from midnight?
4. In an alphanumeric series where A=1, B=2 ... Z=26, find the numerical position value of the 10th term in the alternating letter-number series: A, 2, C, 4, E, 6, ...
5. In a certain code language, if PENCIL is coded as QFODJM, find the numeric equivalent of the encoded word DESK if D=4, E=5, S=19, K=11 are shifted by the same coding pattern.
6. In a certain coding language, if the word MACHINE is coded as 19-3-9-11-15-14-7 based on its alphabetical position shifted by a constant, what is the numerical code for the word DEVICE?
7. Find the missing number in the numerical series: 5, 11, 23, 47, 95, X. Compute the value of X.
8. If a letter series follows a pattern where each consecutive letter advances by a prime number sequence starting from 2 (A + 2, C + 3, F + 5, K + 7), what is the 5th letter in this sequence?
9. In an alphanumeric arrangement where letters correspond to their reverse alphabetical values (Z=1, Y=2...) and numbers retain their face value, compute the sum of codes for letters P, R, and number 45.

Assignment No. 2: BLOOD RELATIONS AND ARRANGEMENT PROBLEMS. DOS: 30/07/2026

1. In a family of 8 members (A, B, C, D, E, F, G, H) across 3 generations, there are 2 married couples. B is the father of D and grandfather of H. C is the mother of G and sister-in-law of E. A is married to B, and F is the brother of D. If the total number of female members in the family is 4, calculate the numerical ratio of male members to female members in the family
2. Six persons (P, Q, R, S, T, U) are sitting in a linear row facing North. P sits second to the left of T. R is not sitting at any extreme end and is adjacent to neither P nor T. Q is sitting immediate right of R. S is at one of the extreme ends. If positions are numbered 1 to 6 from left to right, what is the absolute numerical difference between the position numbers of Q and U
3. In a circular arrangement of 7 friends (A, B, C, D, E, F, G) sitting around a circular table facing the center, A sits third to the left of C and second to the right of E. B is not an immediate neighbor of A or C. D sits second to the left of B. F and G are neighbors. If seats are numbered consecutively from 1 to 7 clockwise starting from A, what is the sum of the seat numbers occupied by B and D?
4. In a family tree analysis, X is the paternal uncle of Y, who is the daughter of Z. W is the mother of X and has 3 children in total with no other siblings. If the total number of male members in this extended family unit is 3 and the total number of female members is 2, calculate the exact numerical product of the total number of male members and female members.
5. In a family of 8 members (A, B, C, D, E, F, G, H) across three generations, there are two married couples. D is the maternal grandfather of H. C is the sister of D. B is married to E, who is the mother of F. G is the son of D and brother of E. If H is the daughter of F, calculate the exact number of female members in the family and determine the relationship of C to H.
6. Six persons P, Q, R, S, T, and U are seated in a linear row facing North. R sits third to the right of P. Q sits at one of the extreme ends. S is the immediate neighbor of Q and sits second to the left of T. If U occupies the remaining seat, calculate the exact absolute linear distance position (1st from left to 6th from left) of U assuming each adjacent seat is separated by 2 meters, starting with position 1 at 0 meters.
7. A family consists of six members P, Q, R, S, T, and U. Q is the son of R, but R is not the mother of Q. P and R are a married couple. S is the brother of P. T is the daughter of Q. U is the sister of T. Compute the total number of male members in this family and find the generational distance (difference in generation levels) between S and T.
8. Five persons A, B, C, D, and E are sitting around a circular table facing the center. A sits second to the left of C. B is not an immediate neighbor of C. D sits second to the left of E. Calculate the exact seating sequence clockwise starting from A, and find the sum of numerical positions if A=1, B=2, C=3, D=4, E=5 in that clockwise arrangement

Assignment No. 3: LOGICAL PUZZLES AND ANALYTICAL REASONING. DOS: 14/08/2026

1. Five projects (P1, P2, P3, P4, P5) are scheduled across 5 consecutive days from Monday to Friday such that exactly one project is scheduled per day. P3 cannot be scheduled on Monday or Friday. P1 must be scheduled exactly two days before P4. P2 is scheduled on Wednesday. If P5 must be scheduled on the day immediately after P4, calculate the numerical day index for P1 (assuming Monday=1, Tuesday=2, ..., Friday=5)
2. Four departments (HR, Finance, IT, Sales) are assigned to 4 distinct floors (1 to 4 from bottom to top). IT is on an even-numbered floor. HR is immediately below IT. Finance is not on floor 1. Sales is on the highest floor where applicable constraints allow. Calculate the numerical sum of the floor numbers assigned to Finance and HR.
3. Six students (S1, S2, S3, S4, S5, and S6) are grouped into two teams of three (Team A and Team B) for a competition based on the following analytical constraints: S1 and S4 cannot be in the same team. S2 must be in Team A. S3 and S5 must be in the same team. If S6 is placed in Team B, calculate the absolute difference between the sum of subscripts of members in Team A and the sum of subscripts of members in Team B
4. In a matching puzzle involving 4 workers (W1, W2, W3, W4) and 4 tasks (T1, T2, T3, T4), W1 cannot perform T1 or T2. W2 must perform T3. W3 cannot perform T4. W4 must perform T1 or T4. If W3 is assigned to T2, calculate the numerical product of the task numbers assigned to W1 and W4
5. In a matching puzzle, four professionals (Doctor, Engineer, Architect, Teacher) named W, X, Y, and Z live in four different cities (Mumbai, Pune, Nagpur, Nashik) not necessarily in that order. W does not live in Mumbai or Pune. The Engineer lives in Nashik. X is the Teacher and lives in Nagpur. Z is the Architect. Determine the exact city where W lives and compute the numerical sum of their alphabetical index values (A=1...Z=26) for the city name associated with W.
6. Five tasks (T1, T2, T3, T4, T5) are to be scheduled sequentially on a single processor. T3 must be scheduled immediately after T1. T4 cannot be scheduled first or last. T2 must precede T5 by exactly two positions (i.e., one task between them). If T1 is scheduled at position 2, calculate the exact valid sequence of tasks and determine the position number of task T5.
7. A group of seven students (A, B, C, D, E, F, G) are to be divided into three distinct project groups: Alpha, Beta, and Gamma, such that every group has at least two members and one group has three members. A and B must be in the same group (Alpha). C and D cannot be in the same group. E must be in Gamma. If C is placed in Beta, calculate the exact distribution of members across Alpha, Beta, and Gamma, and compute the product of the number of members in Beta and Gamma.
8. An analytical scheduling problem requires assigning five shifts (S1, S2, S3, S4, S5) to five days (Monday to Friday). S3 must be assigned on Wednesday. S1 cannot be on Monday or Friday. S4 must be scheduled immediately after S1. S5 is scheduled on Friday. Calculate the exact day assigned to S2 and determine the numerical day index (Monday=1 to Friday=5) for S2.